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The Poetics of Error

User Perception and Artistic Potential of Demographic Misrecognition in AI Advertising

A project report submitted for the degree:

MSc Creative Computing

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Abstract

With the proliferation of Artificial Intelligence Generated Content (AIGC) in the advertising sector, real-time personalized content generation based on facial recognition is reshaping the user experience. However, "Algorithmic Misrecognition" resulting from technical limitations is often dismissed merely as a system failure. Employing a mixed-methods approach ($N = 25$), this study empirically examines the impact of AI accuracy in age and gender recognition on user trust, Ad Skepticism, and Online Purchase Intention.

The study uncovers a privacy paradox: algorithmic accuracy functions as a double-edged sword. On one hand, accurate gender recognition significantly enhances Perceived Personalization and Online Purchase Intention, reflecting the algorithm's instrumental rationality. On the other hand, it simultaneously triggers high levels of Ad Skepticism, as users perceive it as a potential surveillance threat. Conversely, while gender misrecognition mitigates the anxiety of algorithmic omniscience (resulting in lower skepticism) due to the system appearing "clumsy," it leads to a collapse in Perceived Relevance and provokes strong Perceived Intrusiveness. In contrast, the impact of age misrecognition is negligible, often interpreted by users as a "benign glitch."

Drawing on Zach Blas's critique of "Biometric Capture" [1] and Legacy Russell's "Glitch Feminism" [2] , this paper proposes reframing algorithmic errors as a "Generative Glitch." The research indicates that users do not merely pursue accuracy but engage in a dynamic negotiation between the "efficiency derived from accuracy" and the "fear of surveillance." [3]

This finding challenges the "precision is optimal" design dogma. It offers preliminary theoretical and practical insights for utilizing strategic uncertainty to alleviate the anxiety of subjectivity caused by algorithmic omniscience. Furthermore, through the interactive installation *Identity Fragmentor*, this study demonstrates how the algorithmic gaze can be transformed into an artistic space for identity negotiation.

Keywords: AIGC Advertising; Facial Recognition; Algorithmic Misrecognition; Glitch Art; User Trust; Critical Design; Privacy Paradox

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1 Introduction

In the realm of computational advertising, precise ad delivery has long been regarded as the key to enhancing conversion rates. With the deep integration of Large Language Models (LLMs) and computer vision technologies, advertising systems can now not only identify user identities but also generate real-time, "hyper-personalized" content targeting specific demographic characteristics, such as gender and age. While the widespread application of these technologies has improved advertising effectiveness, it has also precipitated numerous controversies and challenges.

On one hand, the "Chilling Effect" [4] triggered by privacy surveillance has become a focal point of public concern. When advertising systems pinpoint users via technologies like facial recognition, users may feel excessively monitored, leading to resistance against the advertisements. On the other hand, inevitable technical errors in algorithmic identification—known as "Algorithmic Misrecognition"—have emerged as a pressing issue. When AI "misreads" a user—for instance, identifying a male as a female, or a young person as elderly—it is not merely a technical glitch but a psychological and cultural event.

Existing advertising research largely focuses on the benefits derived from recognition accuracy, yet rarely explores the impact of misrecognition on user self-perception, emotional experience, and aesthetic sensibility. This "me, but not me" moment poses a critical question: Should it be viewed solely as an error requiring rectification? Or does it inherently contain untapped critical potential and artistic possibility? [5]

This paper aims to answer three core questions through a dual lens of empirical data and art theory:

1. How do demographic recognition errors affect user trust and acceptance of AI advertising?

2. Is there a difference in the psychological weight between gender and age misrecognition?

3. From an artistic perspective, how can this "algorithmic failure" be transformed into a reflective aesthetic experience?

The results of this study challenge the traditional assumption that "accuracy equals trust." Quantitative data reveal a complex privacy paradox: when AI advertising systems attempt to enhance service efficacy by improving recognition accuracy, they trigger a counter-intuitive psychological reaction. Specifically, while high accuracy increases instrumental value (e.g., Online Purchase Intention), it simultaneously reinforces the user's sense of being watched, thereby stimulating defensive psychology (e.g., heightened Ad Skepticism). This leads to a state of contradiction and trade-off in user attitudes.

By integrating empirical data with critical theory, this paper aims to explore how to transform these technical errors from "system defects" into "artistic material," offering ethical guide-

lines for digital identity design in the AI era. Finally, grounded in empirical findings, this paper presents a critical design practice project [6], *Identity Fragmentor*, discussing how to visualize invisible algorithmic judgments through artistic means.

2 Literature Review

2.1 Algorithmic Transparency and the Crisis of Trust

Despite the formidable productivity demonstrated by Artificial Intelligence Generated Content (AIGC) across advertising copy, imagery, and video generation, user trust in these systems remains tenuous. Research indicates that even when systems are explicitly labeled as "AI-generated," the absence of explainable mechanisms or user correction channels can paradoxically exacerbate Ad Skepticism [7]. This opacity precipitates profound concerns as algorithmic technologies are increasingly deployed within the advertising domain. When facial recognition intervenes in the personalization process, this distrust is further amplified by the perception of "dataveillance" [8]. Consequently, users not only question the authenticity of the content but also harbor apprehensions regarding the continuous capture and categorization of their biometric features.

2.2 Facial Recognition in Computational Advertising: The Mechanics of the Gaze

While traditional advertising relies on static user profiles or browsing history, the integration of Facial Recognition Technology (FRT) enables a paradigm shift toward "Real-Time Biometric Profiling" [16]. Technically, these systems operate by mapping facial landmarks—often utilizing 68-point or 128-point feature vectors—to infer demographic attributes such as age, gender, and even emotional states. In the context of "Digital Out-of-Home" (DOOH) advertising, this technology transforms passive screens into active observers, creating a feedback loop where content is dynamically optimized based on the viewer's physical features. However, this "hyper-personalization" relies heavily on the assumption that external biometric features accurately reflect internal consumer needs. As computational power increases, the "algorithmic gaze" becomes more pervasive, raising the stakes for recognition accuracy. When the system fails to map these landmarks correctly—a phenomenon this study examines as "Algorithmic Misrecognition"—it disrupts the seamless commercial narrative, exposing the artificiality of the interaction.

2.3 Algorithmic Bias and Gender Essentialism

AI systems predominantly employ binary classification frameworks in gender recognition tasks, implicitly embodying a form of technological gender essentialism [9]. Such systems not only reduce gender to binary "male/female" labels but often entrench stereotypes within generated content—for instance, associating females with appearance and emotion, while binding males to function and efficiency. This bias not only degrades the user experience but may also constitute potential harm to non-binary or transgender populations. Furthermore, research indicates that algorithmic bias is pervasive in both image recognition and Natural Language Processing (NLP), thereby exacerbating gender inequality.[10]

2.4 The Ontology of Identity: From Representation to Performativity

Underpinning the design of commercial facial recognition systems is a specific ontological assumption: that identity is a pre-existing, objective fact waiting to be "discovered" by the machine. In their meta-analysis of Automatic Gender Recognition (AGR) literature, Keyes (2018) [9] reveals that the vast majority of systems conceptualize gender as immutable, binary, and physiologically discernable. This approach aligns with what Scheuerman et al. (2019) term "Visual Essentialism," where algorithms are tasked with "representation"—that is, truthfully extracting the user's identity from pixel data. [15]

However, this positivist view conflicts with sociological understandings of identity as "performative"—a dynamic construction enacted through social interaction rather than a static biological truth [14] . When an algorithm outputs a binary label (e.g., "Male"), it does not merely describe the user; it actively attempts to construct a digital reality for them. Consequently, algorithmic misrecognition is not merely a technical classification error (a failure of representation) but can be experienced as a form of "ontological violence" that denies the user's agency in defining their own identity. This study challenges the traditional representational paradigm, arguing that the "gap" between algorithmic output and user self-perception is not just noise, but a critical space for identity negotiation.

2.5 Glitch as Art Methodology

In contrast to the engineering perspective, which views "errors" as defects requiring elimination, contemporary art and critical theory ascribe generative significance to the "glitch." In his *Facial Weaponization Suite* [11], Zach Blas posits that biometric technology attempts to incarcerate fluid identities within a "data cage," and rendering the face "illegible" serves as a political strategy of resistance. Similarly, in *Glitch Feminism* [2], Legacy Russell proposes that when non-normative bodies trigger machine errors, this "glitch" is not a failure, but a liberatory moment of refusal against binary coding.

Artists like Refik Anadol, through works such as *Machine Hallucinations* [12], demonstrate the "hallucinations" produced by AI when processing massive datasets, revealing that algorithmic errors can serve as an artistic medium bridging alternative realities and the collective unconscious. These practices collectively point to a distinct possibility: algorithmic "misrecognition" is not merely a manifestation of technical limitations, but an opportunity to unlock new narrative spaces and aesthetic experiences [13].

3 Methodology

3.1 System Design

This study developed a web-based experimental platform integrating computer vision and Large Language Models (LLMs) to achieve real-time personalized advertisement generation. The platform was designed to expose the algorithmic recognition process, thereby stimulating user reflection on recognition accuracy.

- **Frontend Design:** The open-source library *face-api.js* [17] was utilized to capture user faces in real-time and extract age and gender predictions. This library provides efficient facial recognition capabilities, enabling rapid and accurate detection and analysis of facial features.
- **Generation Logic:** Detection results were inserted into a preset Prompt (e.g., "Recommend a product to a 25-year-old woman. Generate lead-generation or conversion-oriented ads designed to guide the user to the next step, such as making a purchase, downloading, registering, or visiting the website.") to invoke an LLM for generating text-only advertising copy. This mechanism ensured the Perceived Personalization and Perceived Relevance of the content.
- **Interaction Design:** The interface explicitly displayed the AI's recognition results (e.g., "Identified as: Female / 25"), forcing users to confront the "algorithmic gaze" regardless of accuracy. This design not only increased transparency but also established a foundation for subsequent user feedback and reflection.

3.2 Experimental Procedure

A total of 25 participants were recruited to ensure sample diversity and representativeness. The experimental procedure was as follows:

1. **User Enables Camera:** Participants enabled the browser camera, and the system generated a personalized advertisement in real-time.

Table 1: Measurement Items and Sources

Category	Question Nos.	Source
Ad Skepticism	Q1-Q9	Baek & Morimoto (2012)
Perceived Personalization	Q10-Q14	Baek & Morimoto (2012)
Perceived Relevance	Q15-Q19	Odoom, (2022)
Online Purchase Intention	Q20-Q22	Odoom, (2022)
Perceived Intrusiveness	Q23-Q27	Odoom, (2022)

2. **Completion of Standardized Scales:** Completion of Standardized Scales: Participants subsequently completed a survey consisting of standardized scales designed to evaluate user reactions and attitudes. As detailed in Table 1, the survey measured Ad Skepticism and Perceived Personalization (adapted from Baek & Morimoto, 2012) [18], as well as Perceived Relevance, Online Purchase Intention, and Perceived Intrusiveness (adapted from Odoom, 2022) [19]. All items were measured on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree).

3. **Evaluation of Recognition Accuracy:** Participants evaluated the accuracy of the AI recognition ("Did it accurately identify your gender/age?"). This step aimed to collect initial user feedback on the algorithmic recognition results.

4. **Semi-Structured Interview:** Finally, participants engaged in a 10–15 minute semi-structured interview to explore their emotional responses and meaning interpretations. The interviews aimed to gain a deeper understanding of user perceptions and experiences regarding algorithmic misrecognition.

3.3 Participants

This study employed convenience sampling to recruit 25 participants ($N = 25$) for a laboratory-controlled experiment. The demographic profile of the sample demonstrated a highly balanced gender distribution, comprising 13 males and 12 females, effectively mitigating the influence of gender bias on algorithmic perception. Participants ranged in age from 20 to 45 years ($M = 30.52$, $SD = 7.89$), covering both digital natives and early digital immigrants, who constitute the primary target audience for generative AI advertising.

Prior to data collection, ethical approval was obtained from the University of the Arts London CCI College Research Ethics Sub-committee. All participants provided informed consent regarding the use of their facial data and were debriefed about the nature of the algorithmic errors post-experiment.

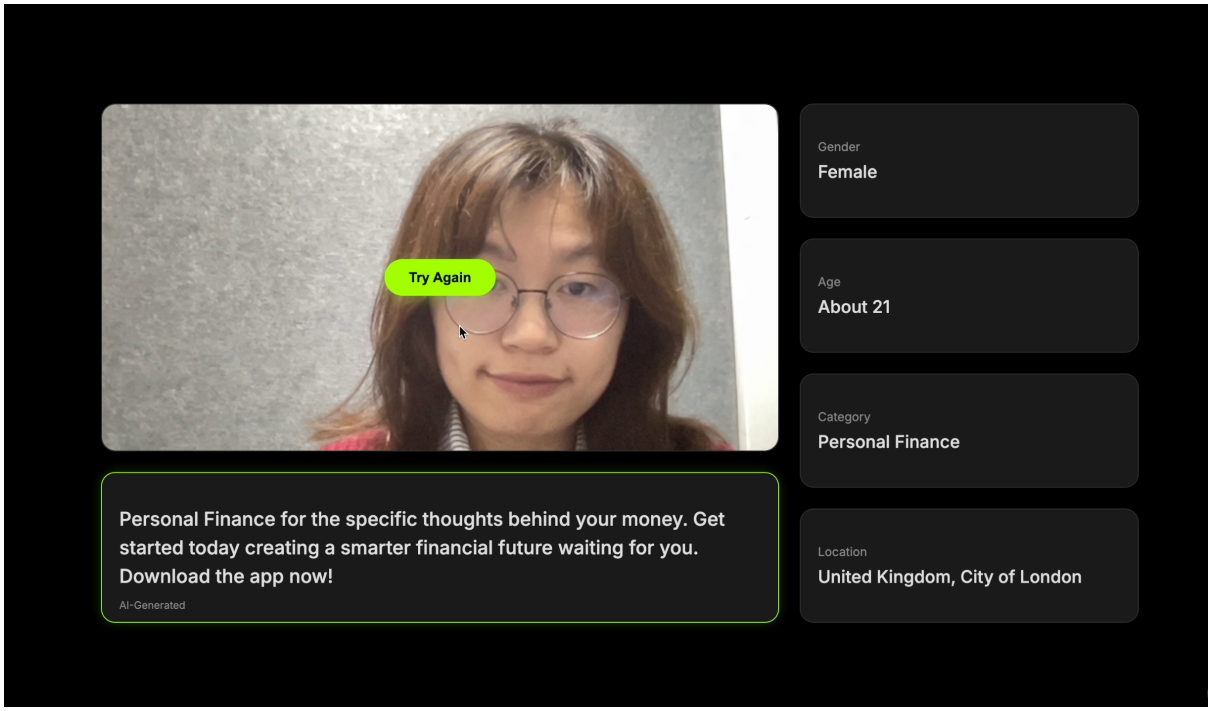


Figure 1: Web-based Experimental Platform

3.4 Analysis Methods

Quantitative: In light of the sample size ($N = 25$) and the characteristics of the cell distribution across groups, this study employed the robust Independent Samples T-test. I examined the main effects of gender accuracy (High vs. Low) and age accuracy (High vs. Low) on five core psychological metrics: Ad Skepticism, Perceived Intrusiveness, Online Purchase Intention, Perceived Personalization, and Perceived Relevance. To rigorously address concerns regarding statistical power associated with small sample sizes, this study incorporated Cohen's d effect size analysis. Adhering to Cohen's (1988) benchmarks [20], I prioritized large effects ($d > 0.8$) to validate the practical significance and robustness of the statistical findings. Furthermore, the study first conducted a rigorous reliability optimization of the measurement tools by calculating Cronbach's α coefficients and Item-Total Correlations. At the level of mechanism exploration, due to sample constraints, I utilized a Pearson Correlation Matrix combined with regression trend analysis to elucidate the potential psychological pathways through which "accuracy" influences Online Purchase Intention.

Qualitative: This study adopted a descriptive interpretative approach to analyze the semi-structured interview data. The analysis followed a three-step process:

- **Pattern Identification:** This study conducted a close reading of the interview transcripts to identify recurrent patterns in users' emotional responses and attitudes toward algorithmic recognition. [23]

- **Theme Extraction:** Based on the key variables (e.g., Ad Skepticism and Perceived Per-

Table 2: Reliability Analysis and Optimization of Measurement Scales

Construct	Items	Initial α	Item Deleted	Final α
Ad Skepticism	Q1 – Q9	0.81	None	**0.81**
Perceived Personalization	Q10 – Q14	0.63	None	**0.63**
Perceived Relevance	Q15 – Q19	0.66	None	**0.66**
Purchase Intention	Q20 – Q22	0.40	Q20 (Purchase Interest)	**0.66**
Perceived Intrusiveness	Q23 – Q27	0.58	None	**0.58**

Note: N=25. Cronbach's $\alpha > 0.60$ is considered acceptable for exploratory research. Item Q20 was removed to improve internal consistency, while other scales retained all items to preserve content validity.

sonalization), this study summarized the participants' feedback into core themes, such as "surveillance anxiety" regarding gender accuracy and "benign errors" regarding age misrecognition.

- **Theoretical Interpretation:** Finally, I selected representative quotes to illustrate these themes and interpreted them through the theoretical lenses of Zach Blas's "Biometric Capture" and Legacy Russell's "Glitch Feminism." This allowed us to discuss the broader implications of algorithmic misrecognition on user subjectivity and aesthetic experience. [22]

3.5 Measurement Tool Revision

User responses were measured using a 5-point Likert scale. Given the exploratory nature of this study ($N = 25$), this study conducted a rigorous reliability test of the data based on the standards established by Hair et al. (2019) [21].

Preliminary analysis indicated that the initial reliability of the Online Purchase Intention scale was low ($\alpha = 0.40$). Screening via Item-Total Correlation analysis revealed that Q20 ("Purchase Interest") demonstrated significantly insufficient consistency with other items. Upon the exclusion of Q20, the Cronbach's α for the scale improved substantially to 0.66, reaching an acceptable level.

Regarding the Perceived Intrusiveness scale, although the initial reliability ($\alpha = 0.58$) was slightly below conventional thresholds, exploratory deletion of any single item failed to yield a significant improvement in reliability (maximum increase of only 0.017). Considering that items Q23–Q27 each covered distinct facets of intrusiveness (e.g., "disturbing," "irritating"), this study decided to retain all items for subsequent analysis to ensure Content Validity. This decision aligns with the priority given to data integrity in exploratory research when sample size is constrained.

4 Results

4.1 Main Effects of Algorithmic Misrecognition: The Cost of Accuracy

First, as a validity check, independent samples t-tests confirmed that accurate gender recognition significantly enhanced users' Perceived Personalization ($M_{High} = 4.26$ vs. $M_{Low} = 3.49, p < .001$) and Perceived Relevance. This demonstrates that the experimental system successfully manipulated the quality of the advertising experience through algorithmic accuracy. To validate the robustness of the findings, this study incorporated effect size analysis. Independent samples t-tests revealed a thought-provoking "Privacy Paradox":

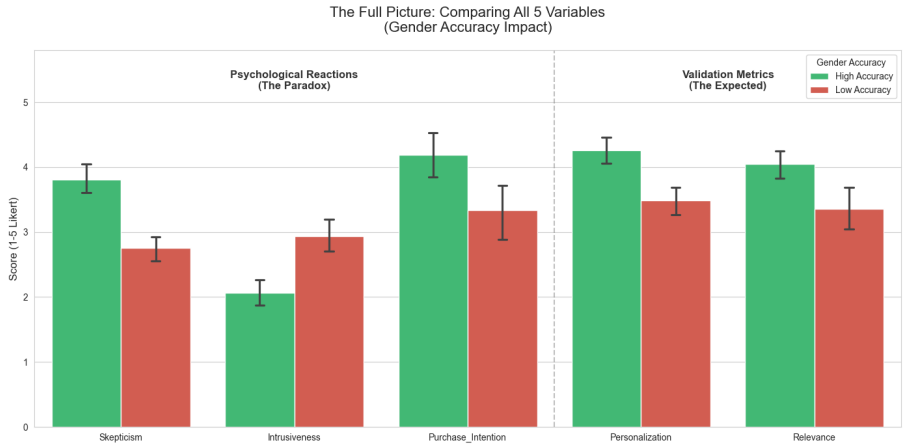


Figure 2: Validity Check

- **The Skepticism Paradox:** Contrary to the original hypothesis, data showed that Accurate Gender Recognition triggered significantly higher Ad Skepticism ($t(23) = -6.26, p < .001, d = -2.61$). Users in the accurate group exhibited extremely high vigilance ($M = 3.81$) [24], whereas misrecognized users conversely displayed higher trust ($M = 2.75$). This suggests that when algorithms precisely capture user gender characteristics, users experience not "service," but a degree of "surveillance," thereby stimulating strong defensive skepticism. [25]

- **Intrusiveness & Misrecognition:** However, regarding the dimension of Perceived Intrusiveness, the trend aligned with traditional expectations [26]: gender misrecognition significantly increased the sense of intrusion ($t(23) = 5.28, p < .001, d = 2.20$). This implies that while misrecognized advertisements are perceived as "honest technical errors" (resulting in lower skepticism), they appear more "disturbing" (higher intrusiveness) due to a lack of Perceived Relevance.

- **Comparison of Effect Sizes:** The impact of gender misrecognition on psychological metrics far exceeded that of age misrecognition. Data indicated that the effect size of the

gender dimension on Ad Skepticism ($d = 2.61$) was 2.4 times that of the age dimension ($d = 1.09$). This confirms that gender, as a core social identity, possesses extremely high sensitivity in algorithmic interactions. This finding resonates with Social Identity Theory (Tajfel & Turner, 1979) [27], suggesting that gender acts as a more salient 'primitive category' in digital self-concept than age, thus eliciting stronger psychological reactance when violated.

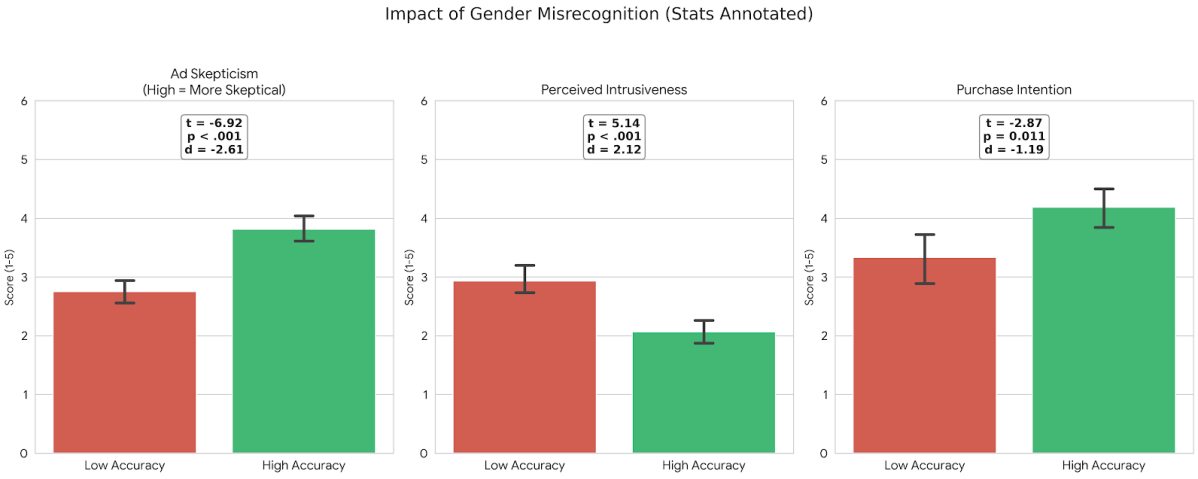


Figure 3: Impact of Gender Misrecognition

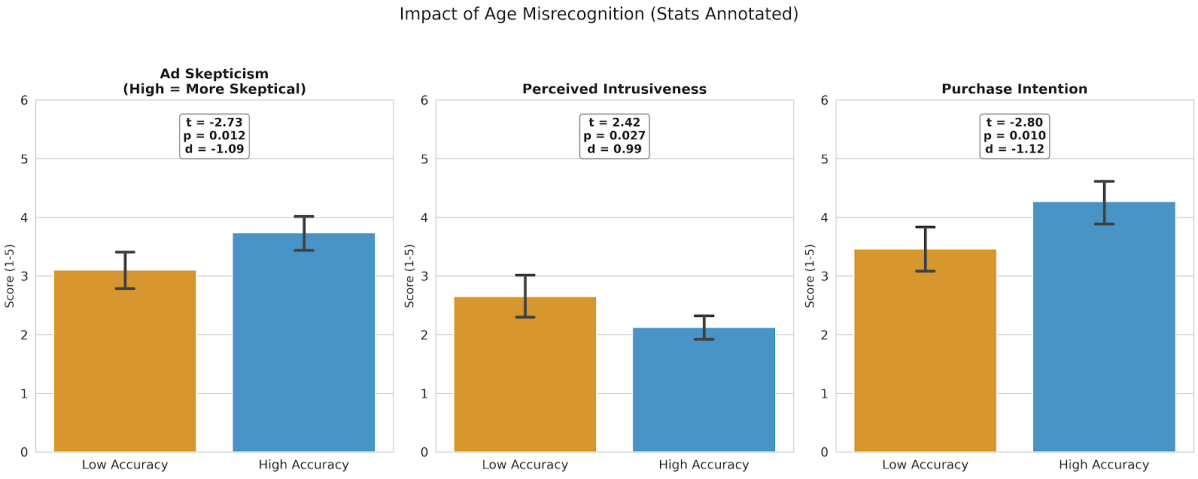


Figure 4: Impact of Age Misrecognition

4.2 The Complex Path of Trust: The Interplay of Correlation and Perceived Threat

Given the exploratory sample size of this study ($N = 25$), conducting a direct causal mediation test may face issues of insufficient statistical power (Baron & Kenny, 1986) [28]. Therefore,

this study analyzed the Pearson Correlation coefficients between variables and the directionality of regression coefficients to elucidate the potential psychological pathways through which "accuracy" influences Online Purchase Intention.

The analysis revealed two distinct and competing psychological mechanisms:

1. **The "Omniscience" Threat Path:** Data indicated a strong positive correlation between gender recognition accuracy and Ad Skepticism ($r = 0.85, p < .001$). This confirms the core hypothesis of the "Privacy Paradox": the more precise the algorithm, the more vigilant the user. Notably, this high skepticism was concurrently associated with higher Online Purchase Intention ($r = 0.69$), suggesting a mindset of "Skeptical Adoption" [29]: while users feel uneasy about algorithmic omniscience, they do not completely reject the service value it provides. [30]

2. **The "Clumsy" Intrusion Path:** On the other hand, gender recognition accuracy showed a significant negative correlation with Perceived Intrusiveness ($r = -0.80, p < .001$). In other words, misrecognition led to a strong sense of intrusion. Regression analysis demonstrated that intrusiveness had a significant negative predictive effect on Online Purchase Intention ($\beta = -0.90, p < .001$), indicating that compared to abstract "skepticism," concrete "intrusiveness" possesses a more direct detrimental impact on purchase decisions.

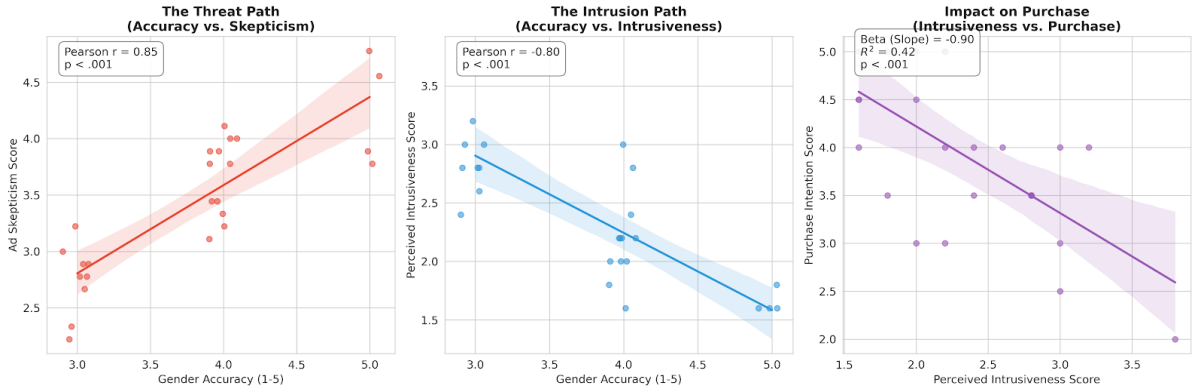


Figure 5: The directionality of regression coefficients

Comprehensive Interpretation: The Dominance of Efficacy Although both "skepticism" and "intrusion" are at play, accuracy itself remains the ultimate determinant of Online Purchase Intention (total effect $\beta = 0.62, p < .01$). This suggests that in the current context of AI interaction, instrumental rationality (efficacy) outweighs privacy anxiety. This aligns with the framework of the Privacy Calculus (Dinev & Hart, 2006) [29], where the utilitarian benefits of accuracy supersede the psychological costs of surveillance.

4.3 Qualitative Findings

The thematic analysis of the semi-structured interviews ($N = 25$) revealed three distinct themes describing how users negotiate with algorithmic errors.

Theme 1: The Uncanny of Identity (Surveillance Anxiety) Participants frequently reported a sense of alienation when the algorithm was too accurate. This phenomenon, which we term "The Uncanny of Identity," refers to the discomfort of being "seen through" by a machine. For example, Participant 12 described the experience:

> "It felt like being assessed by a stranger, almost like a character in a short film. I was curious not just about the ads, but about how the AI perceived me." (P12, Female, 34) Interpreting this through a theoretical lens, this suggests that algorithmic errors transform the system from a neutral tool into an entity with a "gaze," disrupting the user's sense of Perceived Personalization.

Theme 2: Accidental Resonance (The Generative Glitch) In contrast to the anxiety surrounding gender, age misrecognition was often interpreted as a "benign glitch" or even a compliment. This theme, "Accidental Resonance," suggests that errors can foster emotional connection when they align with a user's ideal self-image.

> "When the system identified me as younger than I am, I felt surprisingly connected — it reminded me of my younger self, and I found the ads unexpectedly engaging." (P08, Female, 35) This pattern indicates that such errors function as what Russell (2020) [2] might call a "generative glitch"—an error that fosters emotional connection rather than strictly breaking Perceived Relevance.

Theme 3: Aesthetic Distanciation Finally, several participants shifted their perspective from that of a consumer to that of an art critic. When the system failed spectacularly (e.g., giving nonsensical outputs), users stopped evaluating the ad's relevance and began appreciating the absurdity of the technology.

> "The generated ads have a literary quality. It would feel more poetic if they could directly generate poetry." (P21, Male, 29) This suggests a form of aesthetic distanciation, where the user's focus shifts from Online Purchase Intention to an evaluation of the algorithm's creative potential.

5 Discussion: “Algorithmic Misrecognition” as Art Material

5.1 Gender as a High-Stakes Algorithmic Wager: The Double Bind of Surveillance and Violence

This study confirms that within AI interactions, gender is far from a neutral data point; rather, it constitutes a high-stakes "algorithmic wager." The quantitative results reveal a profound "Privacy Paradox," providing a complex empirical footnote to Zach Blas’s theoretical framework [1].

First, contrary to conventional wisdom, data indicate that accurate gender recognition triggered a skepticism effect size as high as $d = 2.61$. This directly corroborates Blas’s critique of "Biometric Capture": when the algorithm precisely "sees through" the user, this recognition is experienced not as a service, but as a form of "omniscient" surveillance. To be accurately captured is to be locked within the algorithmic gaze, thereby stimulating intense defensive Ad Skepticism.

On the other hand, while gender misrecognition mitigated this anxiety of surveillance (resulting in significantly lower skepticism), it provoked a high level of Perceived Intrusiveness ($d = 2.20$). This psychological backlash corresponds to what Blas terms "Ontological Violence": when the algorithm forcibly locks a fluid body into an erroneous binary category, the nature of the error transcends mere functional failure. It constitutes a symbolic erasure and forced rewriting of the user’s identity.

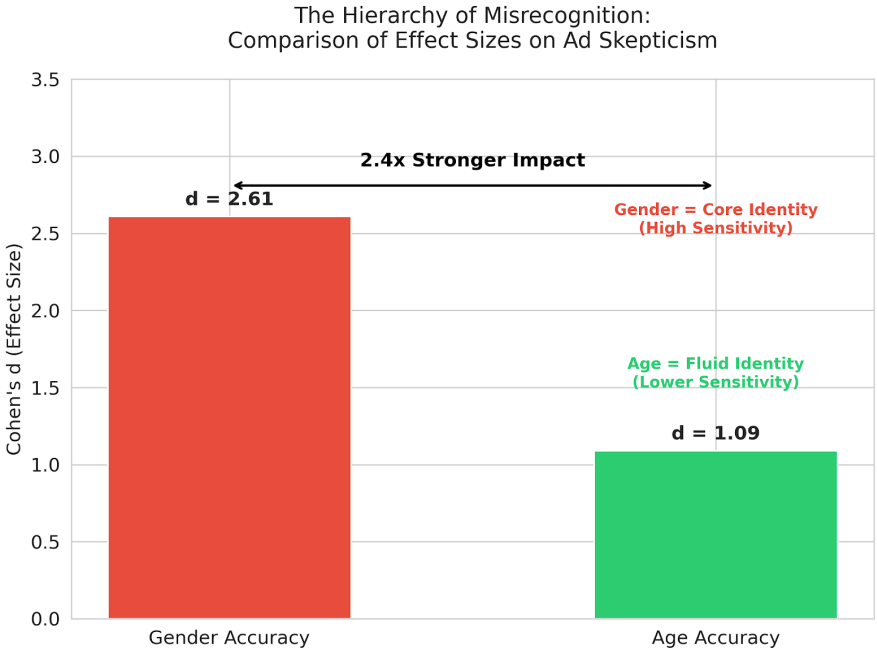


Figure 6: Effect Size Comparison

Consequently, users are effectively trapped in a "Double Bind" [31]: they are either accu-

rately recognized and feel surveilled (High Ad Skepticism), or misrecognized and feel offended (High Perceived Intrusiveness).

5.2 Glitch as Generative Rupture: Age Misrecognition and Identity Negotiation

In contrast to the defensive mechanisms triggered by gender misrecognition, the statistical non-significance of age misrecognition, coupled with interview feedback describing "nostalgia," reveals another dimension of error: the "Generative Rupture."

This forms an intertextual dialogue with Legacy Russell's *Glitch Feminism* [2]: not all errors are negative. Unlike the binary violence of gender, the slippage of age is perceived by users as closer to a "Benign Glitch." When the AI identifies a middle-aged user as a youth, the commercial logic may collapse (low Perceived Relevance), but a narrative space opens up. This rupture creates a "Reflective Pause," [33] allowing users to temporarily suspend normative identity coding and engage in a playful negotiation between the "self" and the "self in the eyes of the algorithm."

Based on this, this study propose a "Hierarchy of Misrecognition" [32]: in critical design, misrecognition involving core social identities (e.g., gender, race) tends to trigger resistance, whereas misrecognition of fluid, non-binary characteristics (e.g., age, mood) is more likely to be transformed into artistic material with aesthetic potential.

5.3 Design Implications: Embracing Uncertainty

Building on these findings, this study propose three future-oriented design principles aimed at transforming algorithmic uncertainty into an interactive resource:

1. **Visible Confidence:** The system should not output absolute judgments (e.g., "You are Male"), but rather present probabilities (e.g., "Male, Confidence 62%"). This approach acknowledges algorithmic limitations, enhancing system honesty and transparency.

2. **Strategic Misrecognition:** n practical applications, designers can develop WebCam-based interactive installations. For instance, when recognition confidence falls below 60%, the system should not force a classification but instead trigger a "Glitch Mode"—generating ambiguous, collage-like poetry or imagery (as suggested by participant P21). This design transforms technical defects into a "blank canvas" inviting user co-creation.

3. **Negotiable Identity:** Providing lightweight feedback entry points (e.g., a "This is not me" button) not only helps improve model accuracy but, more importantly, grants users the right to immediate remediation of their digital identity, shifting the dynamic from passive observation to active negotiation.

6 Critical Design Practice: *Identity Fragmentor*

6.1 Design Philosophy and System Architecture

To translate the theoretical frameworks of the "Generative Glitch" and "Hierarchy of Misrecognition" proposed earlier into a perceptible interactive experience, this study developed an interactive digital art installation [34]—*Identity Fragmentor*. The installation aims to force the audience to visually confront the invisible "Machine Gaze" [35] via visualization techniques.

The installation utilizes real-time facial recognition technology to capture the viewer's biometric features and integrates Generative AI to instantaneously generate two categories of identity labels:

1. **The "Assumed" Subject Identity:** Judgments based on high algorithmic confidence (corresponding to "Accurate Recognition" in the study);
2. **The "Misread" Parallel Fragments:** 20 probability-based, deviating identity hypotheses (corresponding to "Algorithmic Misrecognition" in the study).

These labels float around the viewer's image in a Cyberpunk-style Augmented Reality (AR) formation, constructing a "Datafied Mirror Stage. [36]"

6.2 Empirical Response to Research Findings

The design logic of Identity Fragmentor responds directly to the empirical findings of this study, attempting to dissolve the ontological violence of algorithms through artistic means:

- **Visualizing the "Privacy Paradox":** Quantitative research indicated that accurate recognition triggered high Ad Skepticism. In the installation, eerily accurate labels (e.g., "Precise Social Media Persona") are brutally exposed on the screen, instantiating the user's subconscious sense of Surveillance. This "hyper-visibility" no longer serves commercial monitoring but becomes a medium for reflection, making the audience realize that to be accurately identified is to be locked in.
- **Reconstructing "Misrecognition" as Aesthetic Experience:**Addressing the finding that "gender misrecognition leads to Perceived Intrusiveness," the installation dilutes the aggression of singular errors by pluralizing (generating 20 fragments) and absurdifying (generating "machine hallucinations") the misrecognition. When the screen is flooded with floating tags like "male/30," "none/0," and "unknown/30s," the singular, binary gender violence is deconstructed. Echoing the "aesthetic distanciation" mentioned by interviewees, this approach transforms intimidating technical errors into "floating tags" with narrative potential.

6.3 Interactive Experience: From Definition to Self-Negotiation

The experiential flow of the installation simulates the fluidity of digital identity:

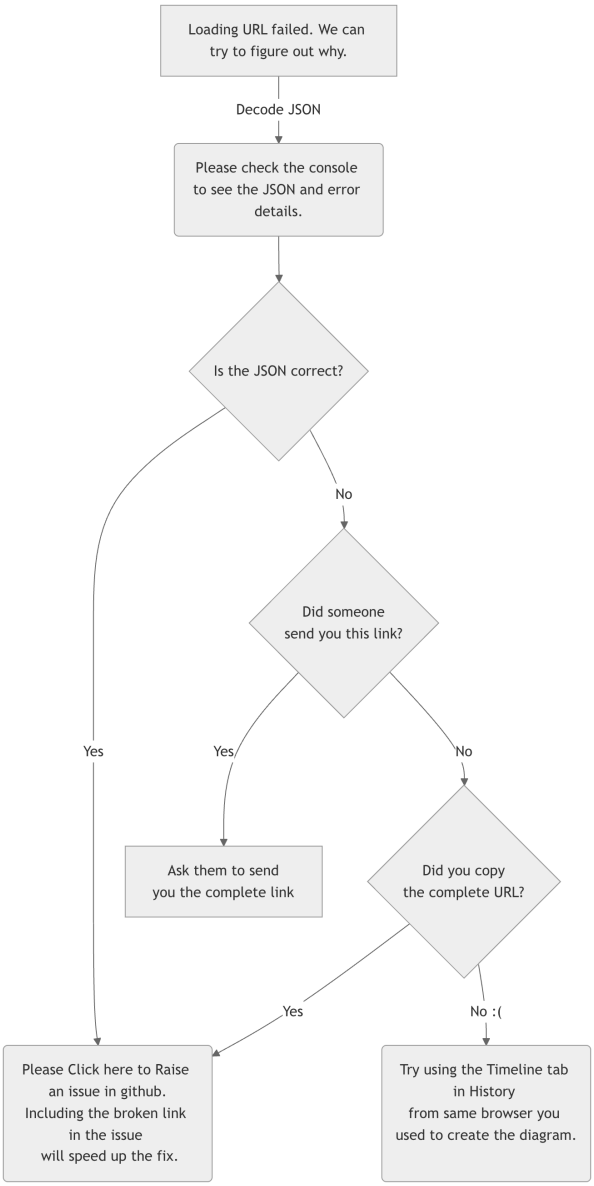


Figure 7: Technical Flowchart

- **The Gaze(Input Phase):** As the viewer approaches the installation, the webcam captures their facial input. Unlike standard mirrors, the system explicitly displays the bounding box and the facial landmarks (68 points), forcing the viewer to see themselves as "data" rather than an image.

- **Fragmentation (Processing Phase):** The system processes the input through two parallel logic paths. The first path outputs the "High Confidence" result (e.g., "Male / 98"). The second path, triggered by a custom "Glitch Function," [5] purposely lowers the con-

fidence threshold to generate 20 divergent, probability-based identity hypotheses (e.g., "Unknown / 45", "Female / 30").

- **Co-existence(Output Phase):**These conflicting labels explode across the screen in an Augmented Reality (AR) overlay. The viewer sees their reflection surrounded by both "accurate" definitions and "hallucinated" possibilities. This visual overload prevents the viewer from being fixed into a single category, effectively creating a "glitch" space that refuses singular coding. At this moment, the screen ceases to be a mirror confirming "Who I am," and becomes a theater showcasing "Who I could be."

Through Identity Fragmentor, this study not only validates the feasibility of "algorithmic misrecognition as artistic material" but also offers a concrete paradigm for critical design: rather than pursuing the (near-impossible) elimination of algorithmic bias, it is more potent to expose bias and noise, thereby returning the right of interpretation to the human subject. [37]

7 Limitations and Future Research

While this study illuminates the psychological mechanisms and aesthetic potential of algorithmic misrecognition, it is imperative to acknowledge several limitations. These constraints not only contextualize the findings but also delineate pathways for future scholarly inquiry:

1. **Sample Limitations and Statistical Power:** The sample size of $N = 25$ constrains the generalizability of statistical inferences. Although effect size analysis (Cohen's d) suggests that the primary findings possess strong internal validity, future research must expand the sample size—particularly by incorporating more diverse racial and age demographics—to corroborate the robustness of these conclusions.

2. **The Paradox of Theory and Method:** While this study theoretically critiques algorithmic gender essentialism, the experimental design—constrained by existing technology (face-api.js)—necessitated the adoption of a binary male/female classification framework [1]. This choice reflects not an endorsement of gender essentialism, but rather the "technical reality" of current mainstream commercial algorithms. Consequently, I was unable to capture the unique experiences of non-binary and transgender populations when facing misrecognition. This "Epistemological Gap" itself serves as a microcosm of current AI limitations; future scholarship should prioritize the development of more inclusive identity labeling systems.

[1]Note: While newer models are beginning to explore non-binary classification, the vast majority of commercially available, lightweight libraries accessible for web-based advertising still rely on binary datasets (Keyes, 2018). [38]

3. **Algorithmic Bias and the "Purity" of Misrecognition:** This study did not independently control for the influence of variables such as skin tone and facial features on algorithmic accuracy. Existing literature indicates that facial recognition algorithms exhibit significantly higher error rates among darker-skinned populations (Buolamwini & Gebru, 2018) [10]. Therefore, instances of "misrecognition" observed herein may be compounded by racial bias rather than constituting purely random technical glitches.

4. **Ecological Validity:** The forced interaction inherent in the laboratory setting may have amplified user attention toward the advertisements. [39] In authentic, fragmented browsing scenarios, users might simply ignore content lacking Perceived Relevance, rather than engaging in deep psychological reflection or exhibiting measurable changes in Ad Skepticism or Online Purchase Intention.

8 Conclusion

Through the intertextual synthesis of empirical evidence and critical theory, this study reexamines the duality of "algorithmic misrecognition" in AI advertising. It answers the core question: Given the inevitability of algorithmic errors, how do we coexist with them?

The core findings reveal a profound paradox in human-machine trust: accuracy is a double-edged sword. While extremely high gender recognition accuracy effectively enhances commercial conversion potential (Online Purchase Intention), it simultaneously triggers a strong user defense mechanism—Ad Skepticism. This data suggests that in the eyes of the user, a system that "knows me too well" often implies it is "too dangerous." Conversely, while gender misrecognition dissipates this threat of omniscient surveillance, it inflicts harm in another dimension—Perceived Intrusiveness, representing a violent falsification of the user's ontological identity.

These findings hold significant practical implications for AI ethics and design. They indicate that the singular pursuit of algorithmic "hyper-precision" is not the ultimate path to establishing user trust; instead, it may precipitate a descent into "Uncanny Valley"-like surveillance anxiety. Qualitative interviews further corroborate this: users crave not cold precision, but an "authenticity" that allows for negotiation.

Consequently, I reposition algorithmic misrecognition as a "Generative Resource." In the dimension of age misrecognition, this study observe the aesthetic potential of the glitch as a narrative rupture; in gender misrecognition, I see the necessity of resisting binary classification violence. The *Identity Fragmentor* installation designed in this study serves as the materialization of this concept, demonstrating that by visualizing the algorithmic gaze, we can transform invisible bias into perceptible opportunities for reflection.

Future AIGC system design should embrace the "Uncertainty Principle" [32]—by displaying confidence probabilities, retaining correction interfaces [40], or even strategically exposing

"clumsiness" to break the hegemony of the algorithmic black box. This approach not only mitigates the user's sense of being surveilled but also preserves breathing room for fluid identity in the digital age.

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A Survey Questionnaire

Survey Questionnaire on Personalized Advertising Experience

Part 1: Core Perceptions

Please indicate your level of agreement with the following statements based on the advertisement you just viewed (Strongly Disagree -> Strongly Agree)

Question	Strongly Disagree -> Strongly Agree				
1. I can depend on getting the truth from this AI-generated personalized ad.	1	2	3	4	5
2. The aim of this AI-generated personalized ad is to inform the consumer.	1	2	3	4	5
3. I believe this AI-generated personalized ad is informative.	1	2	3	4	5
4. This AI-generated personalized ad is generally truthful.	1	2	3	4	5
5. This AI-generated personalized ad is a reliable source of information about the quality and performance of the product.	1	2	3	4	5
6. This AI-generated personalized ad is truth well told.	1	2	3	4	5
7. In general, this AI-generated personalized ad presents a true picture of the product being advertised.	1	2	3	4	5
8. I feel I have been accurately informed after viewing this AI-generated personalized ad.	1	2	3	4	5
9. This AI-generated personalized ad provides consumers with essential information.	1	2	3	4	5
10. This AI-generated personalized ad makes a purchase recommendation that matches my needs.	1	2	3	4	5
11. I think that this AI-generated personalized ad enables me to find products that are tailor-made for me.	1	2	3	4	5
12. Overall, this AI-generated personalized ad is tailored to my situation.	1	2	3	4	5
13. This AI-generated personalized ad makes me feel that I am a unique customer.	1	2	3	4	5
14. I believe that this AI-generated personalized ad is customized to my needs.	1	2	3	4	5
15. I find this AI-generated personalized ad beneficial to me.	1	2	3	4	5
16. This AI-generated personalized ad seems important to me.	1	2	3	4	5
17. I consider this AI-generated personalized ad useful.	1	2	3	4	5
18. The message in this AI-generated personalized ad is meaningful to me.	1	2	3	4	5
19. I find the content of this AI-generated personalized ad personally valuable.	1	2	3	4	5
20. This AI-generated personalized ad makes me interested in making an online purchase.	1	2	3	4	5
21. I have thoughts about buying the advertised product online.	1	2	3	4	5
22. I will probably purchase the advertised product online.	1	2	3	4	5

23. I consider this AI-generated personalized ad intrusive.	1	2	3	4	5
24. I consider this AI-generated personalized ad disturbing.	1	2	3	4	5
25. I consider this AI-generated personalized ad irritating.	1	2	3	4	5
26. I consider this AI-generated personalized ad invasive.	1	2	3	4	5
27. I consider this AI-generated personalized ad uncomfortable.	1	2	3	4	5

Part 2: Accuracy of System Recognition

Question					
How accurate do you think the system's judgment of your gender is? (Very Inaccurate -> Very Accurate)	1	2	3	4	5
How accurate do you think the system's estimation of your age is? (Very Inaccurate -> Very Accurate)	1	2	3	4	5

Part 3: Background Information

Your actual gender: ○ Male ○ Female ○ Other/Prefer not to disclose					
Your actual age: _____ years old					
How interested are you in the displayed ad category [e.g., "sports products"] in daily life? (Not Interested at All -> Very Interested)					

(Optional) Part 4: Open Feedback

Do you have any other opinions, feelings, or concerns about the advertising slogan you just saw, or about this technology that generates advertisements based on facial information? **If yes, please share the reasons behind them.**

B Semi-Structured Interview Guide

Note: These questions served as a guide. The interviewer adapted follow-up probes based on participant responses.

1. Feedback on Ad Content & Experience

- How would you describe the content of the advertisement you just viewed? Did it feel natural or persuasive to you?
- Did the ad feel like it was "made for you," or did it feel generic? Why?

2. Emotional Reaction to Algorithmic Recognition

- The system displayed an estimation of your age and gender. How accurate was this estimation?
- How did you feel when you saw that label?
- If the recognition was incorrect (a "glitch"), did that change your feeling toward the ad? Did you find it funny, annoying, or uncomfortable?

3. Perceptions of Technology & Novelty

- Did this experience feel different from viewing a standard banner ad? In what way?
- Do you have any other opinions, feelings, or concerns about the advertising slogan you just saw, or about this technology that generates advertisements based on facial information?